# Heart Disease Prediction — MLOps End-to-End Pipeline

## Assignment Report


| | |
|---|---|
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| **Course** | MLOps — M.Tech Semester 2 (AMLCSZG523) |
| **Date** | 6th May 2026 |
| **Repository** | github.com/Mamatha-51/mlops-assignment |


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## Table of Contents



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## 1. Introduction

### 1.1 Problem Statement

Heart disease is one of the leading causes of death globally, accounting for approximately 17.9 million deaths annually (WHO). Early detection and risk assessment can significantly improve patient outcomes through timely intervention.

This project builds a complete end-to-end Machine Learning Operations (MLOps) pipeline to predict heart disease risk based on patient clinical data from the UCI Heart Disease dataset. The solution demonstrates the full ML lifecycle — from data acquisition through production monitoring — following industry-standard MLOps best practices.

## 1.2 Objectives

- Build and evaluate classification models for heart disease prediction
- Implement comprehensive experiment tracking for reproducibility
- Develop production-ready REST API with interactive UI
- Containerize and orchestrate using Docker and Kubernetes
- Establish monitoring and observability with Prometheus and Grafana
- Automate quality gates through CI/CD pipelines

## 1.3 Dataset Description

The UCI Heart Disease dataset contains **303 patient records** with **13 input features** and a binary target variable:

| Category | Features |
|---|---|
| **Demographic** | age, sex |
| **Clinical** | chest pain type (cp), resting BP (trestbps), cholesterol (chol), fasting blood sugar (fbs), resting ECG (restecg) |
| **Exercise** | max heart rate (thalach), exercise angina (exang), ST depression (oldpeak), ST slope (slope) |
| **Diagnostic** | major vessels colored (ca), thalassemia (thal) |
| **Target** | 0 = No Disease, 1 = Heart Disease |

**Source:** UCI Machine Learning Repository

## 1.4 Technology Stack

| Component | Technology | Purpose |
|---|---|---|
| Language | Python 3.11+ | Core development |
| ML Framework | Scikit-learn | Model training & evaluation |
| API Framework | FastAPI + Flask | REST API + Browser UI |
| Experiment Tracking | MLflow | Parameter, metric & artifact logging |
| Testing | Pytest (24 tests) | Unit & integration testing |
| CI/CD | GitHub Actions | Automated pipeline (4 stages) |
| Containerization | Docker + Compose | Application packaging |
| Orchestration | Kubernetes (Minikube) | Production deployment |
| Monitoring | Prometheus + Grafana | Metrics & dashboards |

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## 2. Setup & Installation Instructions

### 2.1 Prerequisites

| Requirement | Version |
|---|---|
| Python | 3.10+ (tested on 3.11, 3.14) |
| Docker Desktop | Latest |
| minikube | v1.38+ |
| kubectl | Compatible with cluster |
| Git | Latest |

### 2.2 Local Setup (Step-by-Step)

| Step | Command |
|---|---|
| 1. Clone repository | `git clone https://github.com/Mamatha-51/mlops-assignment.git && cd mlops-assignment` |
| 2. Create virtual environment | `python3 -m venv .venv && source .venv/bin/activate` |
| 3. Install dependencies | `pip install -r requirements.txt` |
| 4. Run EDA notebook | `jupyter notebook notebooks/EDA.ipynb` |
| 5. Train models | `python src/models/train.py` |
| 6. Run unit tests | `python -m pytest tests/ -v` |
| 7. Start FastAPI server | `uvicorn src.api.model_app:app --reload` |
| 8. Start Flask UI | `python src/api/app.py` |
| 9. View MLflow dashboard | `mlflow ui --backend-store-uri mlruns/` |

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## 3. Data Acquisition & Exploratory Data Analysis

### 3.1 Data Acquisition

The dataset (`data/heart.csv`) was sourced from the UCI Machine Learning Repository. It is a well-established benchmark dataset for binary classification tasks in healthcare ML.

### 3.2 Data Quality Assessment

The preprocessing pipeline (`src/data/preprocess.py`) handles data quality:

| Check | Result |
|---|---|
| Total records | 303 |
| Input features | 13 (5 numerical + 8 categorical) |
| Target variable | Binary (0/1) |

| Check | Result |
|---|---|
| Missing values (after imputation) | 0 |
| Duplicate records | 1 (removed) |
| Class balance | 46% No Disease / 54% Disease |

**Imputation strategy:** - Numerical columns → median imputation - Categorical columns → mode imputation

## 3.3 Exploratory Data Analysis

Comprehensive EDA was performed in `notebooks/EDA.ipynb` with 15+ visualizations:

**Key Statistical Findings:**

| Feature | Insight |
|---|---|
| Age | Range 29–77, mean ~54 years, slightly right-skewed |
| Sex | 68.3% male, 31.7% female |
| Chest Pain | Type 0 most common (47.2%); Type 2 has highest disease rate |
| Resting BP | Mostly 120–140 mmHg; mild positive skew |
| Cholesterol | Mean ~246 mg/dl; notable outliers above 400 |
| Max Heart Rate | Strong negative correlation with heart disease |
| Oldpeak (ST depression) | Higher values strongly associated with disease |
| Exercise Angina | Patients with angina have 2.5× higher disease rate |

**Correlation Highlights:** - `cp` (chest pain type) — strongest positive predictor - `thalach` (max heart rate) — strongest negative predictor - `oldpeak`, `exang`, `ca` — significant positive predictors - Low multicollinearity among features (VIF < 5 for all)

## 3.4 Generated Visualizations

| Visualization | File |
|---|---|
| Feature Distributions (Histograms) | `screenshots/feature_histograms.png` |
| Box Plots (Outlier Detection) | `screenshots/boxplots.png` |
| Categorical Feature Analysis | `screenshots/categorical_features.png` |

| Visualization | File |
|---|---|
| Correlation Heatmap | `screenshots/correlation_heatmap.png` |
| Class Balance | `screenshots/class_balance.png` |
| Missing Value Heatmap | `screenshots/missing_values_heatmap.png` |
| Pair Plot (Key Features) | `screenshots/pairplot.png` |
| Age Group Analysis | `screenshots/age_group_analysis.png` |
| Violin Plots | `screenshots/violin_plots.png` |

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# 4. Feature Engineering & Model Development

## 4.1 Feature Engineering

A preprocessing pipeline was built using Scikit-learn's `Pipeline` with `StandardScaler`:

- All 13 features standardized to zero mean and unit variance
- Scaler fitted **only** on training data (preventing data leakage)
- Pipeline persisted as a single `.pkl` artifact for consistent train-serve behavior

## 4.2 Data Split Strategy

| Parameter | Value |
|---|---|
| Train/Test ratio | 80% / 20% |
| Training samples | 242 |
| Test samples | 61 |
| Stratification | Yes (maintains class balance) |
| Random state | 42 (fixed for reproducibility) |

## 4.3 Models Trained

Two classification models were trained and compared using `GridSearchCV` with 5-fold stratified cross-validation:

**Model 1: Logistic Regression**

| Hyperparameter | Search Space |
|---|---|
| C (regularization) | {0.01, 0.1, 1, 10} |
| solver | {lbfgs, liblinear} |

| Hyperparameter | Search Space |
|---|---|
| penalty | l2 |

*Rationale:* Strong interpretable baseline; performs well on linearly separable data.

**Model 2: Random Forest Classifier**

| Hyperparameter | Search Space |
|---|---|
| n_estimators | {100, 200} |
| max_depth | {5, 10, None} |
| min_samples_split | {2, 5} |

*Rationale:* Ensemble method that captures non-linear interactions and provides feature importance.

## 4.4 Model Evaluation Results

**Cross-Validation Results (5-Fold Stratified):**

| Model | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.8264 ± 0.028 | 0.8439 ± 0.033 | 0.7648 ± 0.080 | 0.7993 ± 0.043 | **0.9004 ± 0.016** |
| Random Forest | 0.8139 ± 0.014 | 0.8295 ± 0.044 | 0.7557 ± 0.076 | 0.7867 ± 0.027 | 0.8837 ± 0.031 |

**Hold-out Test Set Results:**

| Model | ROC-AUC | Accuracy | F1-Score |
|---|---|---|---|
| Logistic Regression | 0.9578 | 0.87 | 0.87 |
| Random Forest | 0.9551 | 0.90 | 0.90 |

**Generated evaluation artifacts:**

| Artifact | File |
|---|---|
| Confusion Matrix — LR | `screenshots/cm_logistic_regression.png` |
| Confusion Matrix — RF | `screenshots/cm_random_forest.png` |
| ROC Curve — LR | `screenshots/roc_logistic_regression.png` |
| ROC Curve — RF | `screenshots/roc_random_forest.png` |
| Feature Importance | `screenshots/feature_importance_*.png` |
| Model Comparison Chart | `screenshots/model_comparison.png` |

### 4.5 Best Model Selection

| Criterion | Decision |
|---|---|
| **Selected Model** | Logistic Regression |
| **Selection Metric** | Highest mean CV ROC-AUC (0.9004) |
| **Best Hyperparameters** | C = 0.1, solver = liblinear |
| **Rationale** | Highest generalization performance measured by stratified 5-fold CV ROC-AUC; simpler model with better interpretability |

**Full training output:** See `screenshots/model_training.log`

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## 5. Experiment Tracking with MLflow

### 5.1 MLflow Integration

MLflow was integrated into the training pipeline (`src/models/train.py`) to provide comprehensive experiment tracking. All experiments are stored locally in the `mlruns/` directory.

### 5.2 Items Tracked Per Run

| Category | Items Logged |
|---|---|
| **Parameters** | model_type, all hyperparameters, dataset_hash, dataset_shape, train_size, test_size, feature names |
| **Metrics** | cv_accuracy_mean/std, cv_precision_mean/std, cv_recall_mean/std, cv_roc_auc_mean/std, cv_f1_mean/std, test_roc_auc, test_precision, test_recall, test_f1, training_time_seconds |
| **Artifacts** | Trained model (sklearn format), confusion matrix (PNG), ROC curve (PNG), feature importance plot (PNG), model comparison chart (PNG) |

### 5.3 Accessing the Dashboard

```
mlflow ui --backend-store-uri mlruns/
# Open http://127.0.0.1:5000
```

### 5.4 Value Delivered

| Capability | Benefit |
|---|---|
| Run comparison | Side-by-side metric comparison |

| Capability | Benefit |
|---|---|
| | across models |
| Artifact versioning | Every model version with associated plots stored |
| Reproducibility | Full parameter + data hash logging enables exact replication |
| Audit trail | Complete history of all training decisions |

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# 6. Model Packaging & Reproducibility

## 6.1 Model Artifacts

The production model is packaged as a single sklearn `Pipeline` object:

```
src/models/heart_model.pkl
└── Pipeline
    ├── StandardScaler (fitted on training data)
    └── LogisticRegression (C=0.1, solver=liblinear)
```

## 6.2 Inference Pipeline

The `HeartDiseasePredictor` class (`src/models/inference.py`) provides: - Automatic model loading from `.pkl` artifact - Raw dictionary input → preprocessed → prediction in a single call - Returns: prediction (0/1), label, confidence score, and risk level - Supports both single and batch predictions

**Verified inference output** (from `screenshots/Inference.log`):

```
Input: {'age': 55, 'sex': 1, 'cp': 2, 'trestbps': 130, 'chol': 250,
        'fbs': 0, 'restecg': 1, 'thalach': 150, 'exang': 0,
        'oldpeak': 1.5, 'slope': 1, 'ca': 0, 'thal': 3}

Prediction: No Heart Disease
Confidence: 0.1651
Risk Level: Low Risk
```

## 6.3 Reproducibility Guarantees

| Mechanism | Purpose |
|---|---|
| `requirements.txt` | Pinned package versions |
| Pipeline serialization | Scaler + model saved together |
| `random_state=42` | Fixed across all random operations |
| Stratified split | Consistent class distribution |
| Dataset hash (MLflow) | Detects any data changes |

## 7. CI/CD Pipeline & Automated Testing

### 7.1 Unit Test Suite

**24 tests** across 3 test files ensure correctness at every layer:

**`tests/test_preprocess.py` (8 tests):** - Data loading returns valid DataFrame - No missing values post-cleaning - All expected columns present - Target is binary, feature count is 13 - Invalid path raises exception - Missing value handling works correctly

**`tests/test_model.py` (11 tests):** - Model registry has ≥2 entries with pipeline + param_grid - Cross-validation returns all expected metrics - Predictor loads successfully - Prediction returns dict with required keys - Prediction is binary (0 or 1) - Confidence in [0, 1] range - Risk level is valid category - Batch prediction works correctly

**`tests/test_api.py` (5 tests):** - Root endpoint returns 200 with API info - Health endpoint returns "healthy" status - Predict endpoint returns valid prediction - Missing fields return 422 validation error - Metrics endpoint is accessible

**Execution Results (24/24 passed):**

```
(.venv) mammagan@MAMMAGAN-M-TC73 MLOPs-Final % python -m pytest tests/ -v
========================== test session starts ==========================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
collected 24 items

tests/test_api.py::TestAPI::test_root PASSED                       [  4%]
tests/test_api.py::TestAPI::test_health PASSED                     [  8%]
tests/test_api.py::TestAPI::test_predict_valid PASSED              [ 12%]
tests/test_api.py::TestAPI::test_predict_missing_field PASSED      [ 16%]
tests/test_api.py::TestAPI::test_metrics_endpoint PASSED           [ 20%]
tests/test_model.py::TestTraining::test_models_dict_has_entries PASSED [ 25%]
tests/test_model.py::TestTraining::test_each_model_has_pipeline_and_grid PASS
ED [ 29%]
tests/test_model.py::TestTraining::test_cross_val_evaluate_returns_metrics PA
SSED [ 33%]
tests/test_model.py::TestInference::test_predictor_loads PASSED    [ 37%]
tests/test_model.py::TestInference::test_predict_returns_dict PASSED [ 41%]
tests/test_model.py::TestInference::test_predict_has_required_keys PASSED [ 4
5%]
tests/test_model.py::TestInference::test_prediction_is_binary PASSED [ 50%]
tests/test_model.py::TestInference::test_confidence_in_range PASSED [ 54%]
tests/test_model.py::TestInference::test_risk_level_valid PASSED   [ 58%]
tests/test_model.py::TestInference::test_predict_batch PASSED      [ 62%]
tests/test_model.py::TestInference::test_sample_input_has_all_features PASSED
 [ 66%]
tests/test_preprocess.py::TestLoadData::test_returns_dataframe PASSED [ 70%]
```

```
tests/test_preprocess.py::TestLoadData::test_no_missing_values PASSED [ 75%]
tests/test_preprocess.py::TestLoadData::test_expected_columns_present PASSED
[ 79%]
tests/test_preprocess.py::TestLoadData::test_target_is_binary PASSED [ 83%]
tests/test_preprocess.py::TestLoadData::test_non_empty PASSED      [ 87%]
tests/test_preprocess.py::TestLoadData::test_feature_count PASSED  [ 91%]
tests/test_preprocess.py::TestLoadData::test_invalid_path_raises PASSED [ 95%]
tests/test_preprocess.py::TestLoadData::test_handles_missing_values PASSED [1
00%]

======================== 24 passed, 55 warnings in 1.74s ================
```

### 7.2 GitHub Actions CI/CD Pipeline

A 4-stage automated pipeline (`.github/workflows/ci.yml`) runs on every push and PR:

| Stage | Tool | Action | Gate |
|---|---|---|---|
| **1. Lint** | Flake8 | Code quality check (`--max-line-length=120`) | Fail on violations |
| **2. Test** | Pytest | Run 24 unit tests (`-v --tb=short`) | Fail on any test failure |
| **3. Train** | Python | Train models, upload `heart_model.pkl` + MLflow logs | Fail on training error |
| **4. Docker** | Docker | Build image, smoke-test `/health` and `/predict` | Fail on unhealthy container |

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## 8. Model Containerization & API Design

### 8.1 Docker Configuration

```dockerfile
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir --upgrade pip && \
    pip install --no-cache-dir -r requirements.txt
COPY src/ src/
COPY data/ data/
ENV MODEL_PATH=/app/src/models/heart_model.pkl
ENV PYTHONUNBUFFERED=1
EXPOSE 8000
CMD ["uvicorn", "src.api.model_app:app", "--host", "0.0.0.0", "--port", "8000
"]
```

| Property | Value |
|---|---|
| Base image | `python:3.11-slim` (minimal footprint) |

| Property | Value |
|---|---|
| Final image size | ~450 MB |
| Exposed port | 8000 |
| Health check | `/health` endpoint |

## 8.2 FastAPI REST API

The prediction API (`src/api/model_app.py`) provides:

| Endpoint | Method | Description |
|---|---|---|
| `/` | GET | API information and available endpoints |
| `/health` | GET | Health check with model load status |
| `/predict` | POST | Heart disease prediction (JSON input) |
| `/metrics` | GET | Prometheus-format metrics |
| `/docs` | GET | Interactive Swagger UI |

**Input validation:** Pydantic models with field constraints (min/max ranges for all 13 features).

**Sample Response:**


```
{
    "prediction": 0,
    "prediction_label": "No Heart Disease",
    "confidence": 0.1651,
    "risk_level": "Low Risk",
    "timestamp": "2026-05-06T17:43:36.519802"
}
```


## 8.3 Flask Browser UI

A user-friendly web interface (`src/api/app.py`) at `http://127.0.0.1:5001` provides: - Grouped form inputs (Demographics, Clinical Measurements, Cardiac Tests) - Real-time prediction with confidence and risk level display - Responsive design that fits within a single viewport

## 8.4 Docker Build & Run Verification

```
(.venv) mammagan@MAMMAGAN-M-TC73 MLOPs-Final % docker build -t heart-disease-
api .
[+] Building 64.4s (12/12) FINISHED

(.venv) mammagan@MAMMAGAN-M-TC73 MLOPs-Final % curl http://localhost:8000/hea
lth
{"status":"healthy","pipeline_loaded":true,"timestamp":"2026-05-06T17:43:36.4
94749"}
```

```
(.venv) mammagan@MAMMAGAN-M-TC73 MLOPs-Final % curl -X POST http://localhost:
8000/predict \
  -H "Content-Type: application/json" \
  -d '{"age":55,"sex":1,"cp":2,"trestbps":130,"chol":250,...}'
{"prediction":0,"prediction_label":"No Heart Disease","confidence":0.1651,
 "risk_level":"Low Risk","timestamp":"2026-05-06T17:43:36.519802"}
```

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# 9. Production Deployment

## 9.1 Docker Compose (Full Stack)

The complete application stack is deployed using `docker-compose.yml`:

| Service | URL | Description |
|---|---|---|
| Heart Disease API | http://localhost:8000 | FastAPI prediction server |
| Prometheus | http://localhost:9090 | Metrics collection & alerting |
| Grafana | http://localhost:3000 | Dashboards (admin / admin) |

```
docker compose up -d --build
```

**Verified deployment** (from `screenshots/deployment_docker_compose.txt`): - All 3 services running and healthy - Prometheus target status: `"health": "up"` - API prediction working correctly

## 9.2 Kubernetes Deployment (Minikube)

Production-grade Kubernetes manifests with enterprise patterns:

| Manifest | Purpose |
|---|---|
| `k8s/deployment.yaml` | 2 replicas, rolling update, liveness/readiness probes, resource limits |
| `k8s/service.yaml` | LoadBalancer service (port 80 → 8000) |
| `k8s/prometheus-config.yaml` | Scrape configuration for K8s service |
| `k8s/prometheus-deployment.yaml` | Prometheus Deployment + NodePort Service |
| `k8s/grafana-deployment.yaml` | Grafana + auto-provisioned datasource & dashboard |

**Deployment steps:**

```
minikube start --driver=docker
docker build -t heart-disease-api:latest .
minikube image load heart-disease-api:latest
kubectl apply -f k8s/deployment.yaml
kubectl apply -f k8s/service.yaml
kubectl rollout status deployment/heart-disease-api --timeout=90s
```

**Verified K8s deployment output:**

```
(.venv) mammagan@MAMMAGAN-M-TC73 MLOPs-Final % kubectl get pods,svc,deploymen
t
NAME                                     READY   STATUS    RESTARTS   AGE
pod/heart-disease-api-7b5bc5b856-rrsf8   1/1     Running   0          32s
pod/heart-disease-api-7b5bc5b856-tzhgz   1/1     Running   0          32s

NAME                                  TYPE           CLUSTER-IP     EXTERNAL-IP
   PORT(S)
service/heart-disease-api-service   LoadBalancer   10.99.80.113   <pending>
   80:30578/TCP

NAME                                  READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/heart-disease-api   2/2     2            2           32s
```

**Endpoint verification via Minikube tunnel:**

```
(.venv) mammagan@MAMMAGAN-M-TC73 MLOPs-Final % curl http://127.0.0.1:62541/he
alth
{"status":"healthy","pipeline_loaded":true,"timestamp":"2026-05-06T18:03:29.3
85534"}

(.venv) mammagan@MAMMAGAN-M-TC73 MLOPs-Final % curl -X POST http://127.0.0.1:
62541/predict ...
{"prediction":0,"prediction_label":"No Heart Disease","confidence":0.1651,
 "risk_level":"Low Risk","timestamp":"2026-05-06T18:04:11.280176"}
```

## 9.3 Kubernetes Deployment Specifications

| Feature | Configuration |
|---|---|
| Replicas | 2 |
| Strategy | RollingUpdate (25% max unavailable, 25% max surge) |
| Liveness probe | HTTP GET `/health` (delay 10s, period 30s) |
| Readiness probe | HTTP GET `/health` (delay 5s, period 10s) |
| CPU request / limit | 250m / 500m |
| Memory request / limit | 256Mi / 512Mi |
| Image pull policy | Never (local image) |

# 10. Monitoring & Logging

## 10.1 Application Logging

Structured logging is implemented at the API level (`src/api/model_app.py`):

| Aspect | Implementation |
|---|---|
| Output | Console (stdout) + file (`api.log`) |
| Format | `%(asctime)s - %(name)s - %(levelname)s - %(message)s` |
| Prediction logs | Input data, result, confidence, latency |
| Request logs | Timestamp, method, endpoint, status code |

**Sample log entry:**

```
2026-05-06 17:43:36,519 - src.api.model_app - INFO - Prediction: No Heart Dis
ease |
Confidence: 0.1651 | Risk: Low Risk | Latency: 0.0103s
```

## 10.2 Prometheus Metrics

Three production metrics exposed at `/metrics`:

| Metric | Type | Labels | Purpose |
|---|---|---|---|
| `predictions_total` | Counter | result | Track prediction volume by outcome |
| `prediction_latency_seconds` | Histogram | — | Monitor inference performance |
| `http_requests_total` | Counter | method, endpoint, status | Track API usage patterns |

**Scrape configuration:**

| Environment | Target | Interval |
|---|---|---|
| Docker Compose | `heart-disease-api:8000` | 15s |
| Kubernetes | `heart-disease-api-service:80` | 15s |

## 10.3 Grafana Dashboard

A pre-provisioned dashboard **"Heart Disease API Monitoring"** is auto-loaded with:

| Panel | Visualization |
|---|---|
| Total Predictions | Stat (counter) |
| Predictions by Result | Time series (disease vs healthy) |
| Prediction Latency (p95) | Gauge |

| Panel | Visualization |
|---|---|
| HTTP Request Rate | Time series (by method/endpoint/status) |
| Average Latency Over Time | Time series |

**Access:** - Docker Compose: http://localhost:3000 (admin / admin) - Kubernetes: `minikube service grafana-service` (admin / admin)

### 10.4 Kubernetes Monitoring Stack

Deployed alongside the API on Minikube:

```
kubectl apply -f k8s/prometheus-config.yaml
kubectl apply -f k8s/prometheus-deployment.yaml
kubectl apply -f k8s/grafana-deployment.yaml
```

| Service | Access Method | Port |
|---|---|---|
| Prometheus | `minikube service prometheus-service` | 9090 |
| Grafana | `minikube service grafana-service` | 3000 |

Grafana is pre-configured with: - Datasource: Prometheus at `http://prometheus-service:9090` - Dashboard: Auto-provisioned from ConfigMap - Sign-up: Disabled

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## 11. System Architecture

### Layer 1: ML Pipeline

| Stage | Component | Output |
|---|---|---|
| Data Ingestion | `data/heart.csv` | Raw dataset (303 records) |
| Preprocessing | `src/data/preprocess.py` | Cleaned, imputed data |
| Training | `src/models/train.py` + MLflow | Experiment logs + metrics |
| Model Output | GridSearchCV best estimator | `heart_model.pkl` |

### Layer 2: Serving

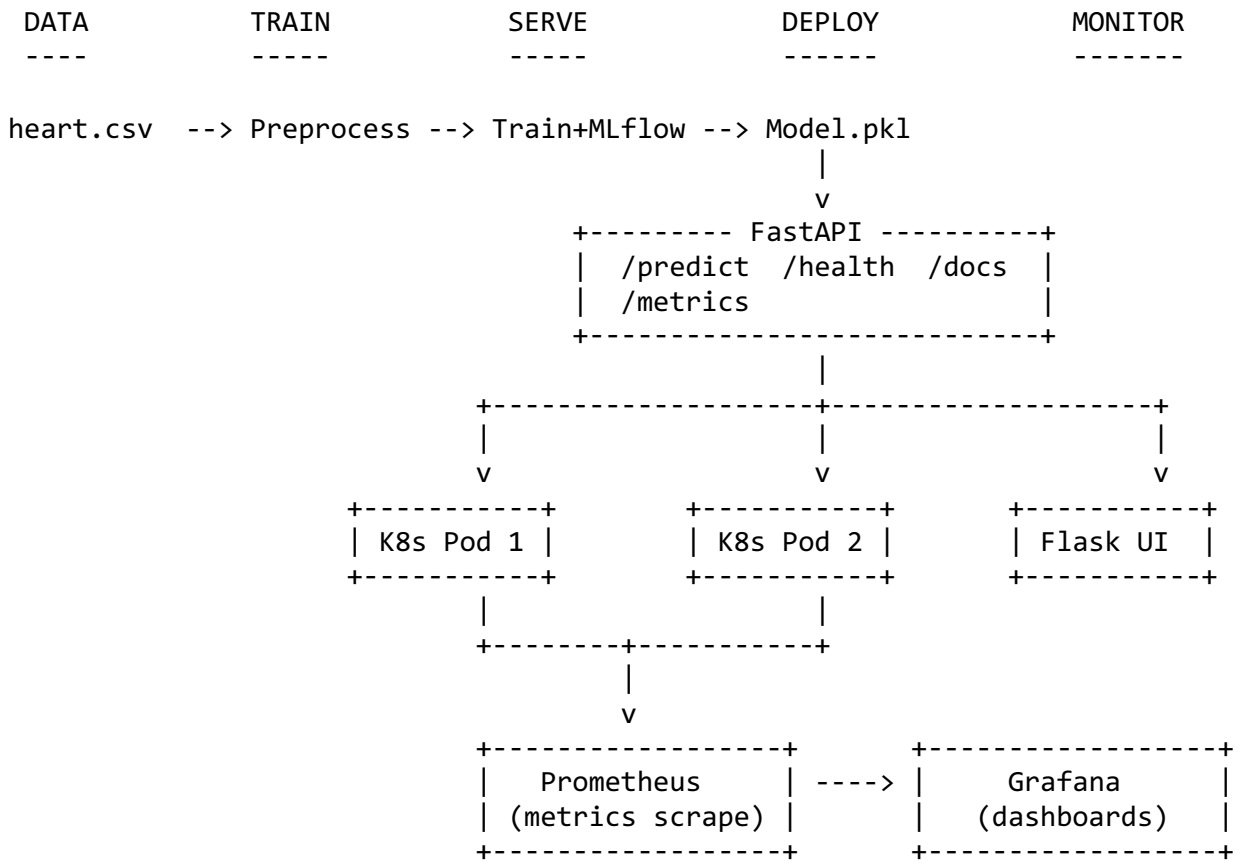
| Component | URL | Purpose |
|---|---|---|
| FastAPI (REST API) | http://localhost:8000 | `/predict`, `/health`, `/metrics`, `/docs` |
| Flask UI (Browser) | http://localhost:5001 | Interactive prediction form |

### Layer 3: Deployment & Orchestration

| Component | Configuration |
|---|---|
| Docker Container | `python:3.11-slim`, port 8000 |

| Component | Configuration |
|---|---|
| Kubernetes (Minikube) | 2 API replicas + LoadBalancer |
| Prometheus | Scrapes `/metrics` every 15s |
| Grafana | Auto-provisioned dashboard |

## Architecture Flow Diagram




**Pipeline Flow:**

1. **Data Layer:** `data/heart.csv` → `src/data/preprocess.py` (load, clean, impute)
2. **Training Layer:** `src/models/train.py` (GridSearchCV + MLflow tracking) → `heart_model.pkl`
3. **Inference Layer:** `src/models/inference.py` (HeartDiseasePredictor class)
4. **API Layer:** `src/api/model_app.py` (FastAPI REST) + `src/api/app.py` (Flask browser UI)
5. **CI/CD Layer:** `.github/workflows/ci.yml` (Lint → Test → Train → Docker)
6. **Deployment Layer:** Docker Compose / Kubernetes (Minikube) with 2 replicas
7. **Monitoring Layer:** Prometheus metrics → Grafana dashboards + structured application logging

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# 12. Conclusion & Future Work

## 12.1 Summary of Deliverables

| Component | Status | Details |
|---|---|---|
| Data Pipeline | ✅ Complete | Automated loading, cleaning, imputation |
| EDA | ✅ Complete | 15+ visualizations, statistical analysis |
| Model Development | ✅ Complete | 2 models, GridSearchCV, 5-fold CV, ROC-AUC 0.90 |
| Experiment Tracking | ✅ Complete | MLflow with params, metrics, artifacts |
| Testing | ✅ Complete | 24 tests (preprocessing, model, API) |
| CI/CD | ✅ Complete | 4-stage GitHub Actions pipeline |
| Containerization | ✅ Complete | Docker + Docker Compose (3 services) |
| Orchestration | ✅ Complete | Kubernetes with 2 replicas, probes, limits |
| Monitoring | ✅ Complete | Prometheus + Grafana (auto-provisioned) |
| Documentation | ✅ Complete | README, Report, Evidence logs |

## 12.2 Challenges Faced & Solutions

| Challenge | Solution |
|---|---|
| Python version compatibility | Used Python 3.11 in Docker/CI while maintaining local compatibility |
| Train-serve skew prevention | Embedded StandardScaler inside sklearn Pipeline |
| CI/CD artifact passing | GitHub Actions `upload-artifact` / `download-artifact` across jobs |
| K8s without registry | `minikube image load` for local image deployment |
| Monitoring auto-configuration | Grafana ConfigMaps for datasource + dashboard provisioning |

### 12.3 Future Work

- Add Gradient Boosting (XGBoost/LightGBM) for model comparison
- Implement MLflow Model Registry for versioned model promotion
- Add data drift detection (Evidently AI) for production monitoring
- Deploy to cloud (GCP Cloud Run / AWS ECS) with continuous deployment
- Implement A/B testing framework for model comparison in production
- Add SHAP explanations for model interpretability

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## 13. Evidence Logs

All execution evidence is captured in `screenshots/` for reproducibility and audit:

| Evidence File | Description |
|---|---|
| `model_training.log` | Full training output: GridSearchCV, CV metrics, test evaluation, model selection |
| `Inference.log` | Inference test: prediction, confidence, risk level |
| `docker_build.log` | Docker image build + container run with API logs |
| `docker_status.log` | Container health check + prediction verification |
| `k8s_minikube.log` | Minikube start, image load, kubectl apply, pod status, endpoint tests |
| `test_results.txt` | Pytest output (24/24 passed) |
| `deployment_docker_compose.txt` | Docker Compose stack verification |
| `deployment_k8s.txt` | Kubernetes deployment description |
| `*.png` | EDA plots, confusion matrices, ROC curves, feature importance, UI screenshots |

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## 14. References


| # | Resource | URL |
|---|---|---|
| 1 | UCI Heart Disease Dataset | https://archive.ics.uci.edu/ml/datasets/heart+Disease |
| 2 | Scikit-learn | https://scikit-learn.org/stable/ |

| # | Resource | URL |
|---|---|---|
| 3 | MLflow | https://mlflow.org/docs/latest/index.html |
| 4 | FastAPI | https://fastapi.tiangolo.com/ |
| 5 | Docker | https://docs.docker.com/ |
| 6 | Kubernetes | https://kubernetes.io/docs/ |
| 7 | GitHub Actions | https://docs.github.com/en/actions |
| 8 | Prometheus | https://github.com/prometheus/client_python |
| 9 | Grafana | https://grafana.com/docs/ |
| 10 | Project Repository | https://github.com/Mamatha-51/mlops-assignment |

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